NAZARBAYEV UNIVERSITY

Course Specification Report

CSCI 423 - Introduction to Parallel Systems and GPU Programming
Параллельді жүйелерге кіріспе және графикалық процессорларды бағдарламалау
Введение в параллельные системы и программирование графических процессоров

General Course Information

School: School of Engineering

Course Type: Elective Course

Semester in which the course is offered: Fall & Spring

Course instructor(s): Ben Tyler

Talgat Manglayev

Course Description - English

This course is intended for students interested in the efficient use of modern parallel systems ranging from mobile multi-core to server/cluster many-core systems. Topics such as parallel computer architectures, the programming models, memory hierarchy, parallel program design and parallel programming tools for multi-core systems and general-purpose graphics processing units (GPGPUs) will be covered. In the second part, the course covers the most common and current GPU parallel programming techniques with lab-based programming assignments using CUDA API

Course Description - Kazakh

Бұл курс мобильдіден сервер/кластерге дейін көп ядролы жүйелердегі қазіргі заманғы параллель жүйелерді тиімді пайдалануға қызығатын студенттерге арналған. Параллельді компьютер архитектурасы, бағдарламалау модельдері, жад иерархиясы, параллельді бағдарламаларды жобалау және көп ядролы жүйелер мен жалпы мақсаттағы графикалық процессорлар (GPGPU) үшін параллель бағдарламалау құралдары сияқты тақырыптар қарастырылады. Курстың екінші бөлімі CUDA API арқылы зертханалық бағдарламалау тапсырмалары бар GPU параллельді бағдарламалаудың ең кең таралған және ағымдағы әдістерін қамтиды.

Course Description - Russian

Этот курс предназначен для студентов, интересующихся эффективным использованием современных параллельных систем, начиная от многоядерных мобильных устройств и заканчивая многоядерными серверными/кластерными системами. Будут рассмотрены такие темы, как архитектура параллельных компьютеров, модели программирования, иерархия памяти, проектирование параллельных программ и инструменты параллельного программирования для многоядерных систем и графических процессоров общего назначения (GPGPU). Во второй части курса рассматриваются наиболее распространенные и актуальные техники параллельного программирования на GPU с лабораторными заданиями по программированию, использующими API CUDA.

Course Requisites

Conditions of Enrollment

Pre-requisites (if applicable): CSCI 332 (C- and above)

Anti-requisites (if applicable): PHYS 421

Course Aims

1. To introduce students to concepts, hardware architectures and software programming models of parallel systems
2. To facilitate students' knowledge and understanding of parallel programming technologies
3. To provide students hands-on experience in designing and implementing simple parallel programs

Course Learning Outcomes (CLOs)

Course Learning Outcomes	Graduate Attributes
1. Define concepts related to CPU and GPU hardware architectures from mobile multi-core to server/cluster many-core parallel systems.	**GA1** - Possess an in-depth and sophisticated understanding of their domain of study. **GA2** - Be intellectually agile, curious, creative and open-minded. **GA3** - Be thoughtful decision makers who know how to involve others. **GA4** - Be entrepreneurial, self-propelling and able to create new opportunities. **GA8** - Be prepared to take a leading role in the development of their country.
2. Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	**GA1** - Possess an in-depth and sophisticated understanding of their domain of study. **GA2** - Be intellectually agile, curious, creative and open-minded.
3. Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks	**GA4** - Be entrepreneurial, self-propelling and able to create new opportunities. **GA5** - Be fluent and nuanced communicators across languages and cultures. **GA6** - Be cultured and tolerant citizens of the world.

	GA8 - Be prepared to take a leading role in the development of their country.
4. Design and implement parallel programs using the parallel programming techniques and CUDA	**GA3** - Be thoughtful decision makers who know how to involve others. **GA4** - Be entrepreneurial, self-propelling and able to create new opportunities. **GA6** - Be cultured and tolerant citizens of the world. **GA7** - Demonstrate high personal integrity. **GA8** - Be prepared to take a leading role in the development of their country.
5. Integrate current research findings, methodologies, and advancements into heterogeneous computing	**GA3** - Be thoughtful decision makers who know how to involve others. **GA5** - Be fluent and nuanced communicators across languages and cultures.

Mapping of Major Core and Elective Courses

Program: Bachelor of Science in Computer Science (Historical)

Course Learning Outcomes	Program Learning Outcomes
1. Define concepts related to CPU and GPU hardware architectures from mobile multi-core to server/cluster many-core parallel systems.	1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. 2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
2. Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. 2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks	1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. 2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline. 6. Apply computer science theory and software development fundamentals to produce computing-based solutions.
4. Design and implement parallel programs using the parallel programming techniques and CUDA	1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. 2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline. 6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

5. Integrate current research findings, methodologies, and advancements into heterogeneous computing	1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. 2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline. 6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

ECTS and Student Workload

Credits: 6

Contact Hours: 45

Self-Study Hours: 105

Total Hours: 150

Assessment Methods

Assessment Methods	Course Learning Outcomes (Reference #)
Lab and Homework Assignments	1, 2, 3, 4, 5
Quizzes	1, 2, 3, 4, 5
Midterm Exams (Two)	1, 2, 3, 4, 5

Assessment Methods Weighting

Assessment Methods	Weighting
Lab and Homework Assignments	30%
Quizzes	30%
Midterm Exams (Two)	40%
Total weighting	**100%**

Course Outline

Session Number	Theme (Key Topic)	Course Learning Outcomes (Reference #)

1	Course Overview. Introduction to parallel systems	1, 2
2	Fundamentals of Accelerated Computing with CUDA C/C++	2, 3, 4, 5
3	Fundamentals of Accelerated Computing with Modern CUDA C++	2, 3, 4, 5
4	Fundamentals of Accelerated Computing with Modern CUDA C++	2, 3, 4, 5
5	Optimize CUDA programs using Nsight profiling tools.	1, 2, 3, 4, 5
6	CUDA C++ features. Reduction algorithm implementation variations using CUDA C/C++.	1, 2, 3, 4
7	Review and Midterm	1, 2, 3, 4, 5
8	Fundamentals of accelerated computing with CUDA python. Introduction to CUDA with Numba. Custom CUDA Kernels in Python with Numba.	1, 2, 3, 4
9	Effective Use of the Memory Subsystem. Nsight Analysis System: Build Custom Python Analysis Scripts.	1, 2, 3, 4
10	Fundamentals of Accelerated Data Science.	1, 2, 3, 4
11	Fundamentals of Accelerated Data Science.	3, 4
12	Fundamentals of Deep Learning with Python calling CUDA functions.	1, 2, 3, 4
13	Fundamentals of Deep Learning with Python calling CUDA functions.	1, 2, 3, 4
14	Research issues on recent parallel systems. Summary and Course Review	1, 2, 3, 4

Learning and Teaching Methods

Course Learning Outcomes	Learning and Teaching Methods
1. Define concepts related to CPU and GPU hardware architectures from mobile multi-core to server/cluster many-core parallel systems.	Laboratories Homeworks; Quizzes; Exams
2. Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	Laboratories Homeworks; Quizzes; Exams
3. Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks	Laboratories Homeworks; Quizzes; Exams

4. Design and implement parallel programs using the parallel programming techniques and CUDA	Laboratories Homeworks; Quizzes; Exams
5. Integrate current research findings, methodologies, and advancements into heterogeneous computing	Laboratories Homeworks; Quizzes; Exams

Laboratory Content

During labs, students are asked to solve tasks relevant to a new topic presented to them. The tasks have prepared code and students need to solve the remaining part of the task. Students are invited one by one to the teacher’s PC, which is projected to the class, to solve a task. The other students may either work on the task independently or follow the selected student’s solution. After completing the task, the student explains their approach to the class.

The students complete approximately one NVIDIA workshop every two weeks, where any remaining tasks are completed at home. In this way, students can earn a certificate of completion from NVIDIA, which they need to upload to receive a grade for their lab.

E-Learning

The course is not designed as an online course, but will make use of some online materials.

Online platform learning.nvidia.com Deep Learning Institute of NVIDIA. The online courses are step-by-step tutorials which have tasks at the end of each course.

Grading

Undergraduate

Letter Grade	Percent Range	Grade (quality points)	Grade Descriptors
A	95-100	4.00	Student has exceeded expectations for all LOs, with evidence of highly comprehensive and in-depth knowledge, understanding and skills
A-	90-94.9	3.67	
B+	85-89.9	3.33	Student has achieved all the LOs of the assessment, exceeding expectations for some LOs, with evidence of thorough knowledge, understanding and skills.
B	80-84.9	3.00	
B-	75-79.9	2.67	
C+	70-74.9	2.33	Student has achieved all the LOs of the assessment with evidence of satisfactory knowledge, understanding and skills.
C	65-69.9	2.00	
C-	60-64.9	1.67	
D+	55-59.9	1.33	Student has partially achieved the LOs of the assessment with evidence of only limited knowledge, understanding and skills.
D	50-54.9	1.00	

Letter Grade	Percent Range	Grade (quality points)	Grade Descriptors
F	0-49.9	0	Student has failed to achieve the LOs of the assessment with very little or no evidence of knowledge, understanding and skills.

Key Books and Learning Materials

Textbook

#	Author(s):	Title	Publisher	Year
1	Wen-mei W. Hwu, David B. Kirk and Izzat El Hajj	Programming Massively Parallel Processors: A Hands-on Approach	Morgan Kaufmann	2026
2	Jason Sanders, Kandrot Edward and Jack Dongarra	CUDA by Example	Upper Saddle River, N.J: Addison-Wesley	2011
3	John L. Hennessy and David A. Patterson	Computer Organization and Design 4th Edition (Ch. 7)	MK Publications	
4	John L. Hennessy and David A. Patterson	Computer Architecture: A Quantitative Approach 5th Edition (Ch. 3 and Ch. 4)	MK Publications	
5		CUDA C Programming Guide (E-resources)		

Software

#	Title	Years and Edition	Publisher/ Manufacturer	Description
1	Nvidia CUDA SDK on Linux			

Academic Integrity Statement

Student Code of Conduct and Disciplinary Procedures

While general guidelines regarding academic integrity are published in the NU Student Handbook and by the school, for this course you must adhere to the following rules:

- You are not allowed to distribute or indirectly make your work available to others, even if you are not intending them to copy it -- this is still considered academic misconduct.
- While it is okay to talk through high-level concepts and ideas with your fellow students and to study with them, you are not allowed to give or receive direct help from others on graded assignments.
- Using code that you find online is strictly forbidden, as is using code generation tools such as ChatGPT, MS Copilot, Github Copilot, etc.
- Any appearance of cheating or inappropriate copying should be avoided, as such instances will be investigated and reported to the school.

- You may only get help on graded work from designated people—the instructors or TAs for the course. If you are struggling with something, by all means, please seek help from them.

In the event that academic misconduct such as plagiarism or cheating is discovered, the student will receive no credit for the work, and the event reported to your school. Egregious cases, or a second offense, can result in failure of the course and potential suspension or expulsion from the university.

When a student suspects that another student has violated the academic honesty policy, a report should be made to the appropriate faculty member.

Course Expectations

The Nazarbayev University Special Learning Needs Committee (SLNC) is committed to creating an equitable and inclusive education environment for all students. If you have a qualified special learning need (physical, cognitive, socio-emotional, and psychological), please contact the SLNC as early as possible to ensure you receive the fullest support available.

For more information, please contact us at SLNC@nu.edu.kz and follow this link to the Accommodation Request Instructions.

Course Expectations

Attendance

Missing classes and habitual tardiness will have a negative effect on your grade, both directly and indirectly. You are also responsible for any announcements made during the class period, so be sure to ask your instructor, TAs, or classmates for any info that you may have missed if you did not attend.

Electronic Resources

Labs will be conducted in one of our hybrid computer labs, which are designed to accommodate the full range of course activities. The necessary programming tools are installed on the classroom lab computers, and are available online for free download to your own computer. You are expected to check your Nazarbayev University e-mail on a daily basis for updates and announcements about the course.

Assignment Submission and Late Policy

Assignments must be submitted by the announced due date and time, as directed by the instructor. Some assignments may need to be submitted in the form of physical hard-copy in class, generally within the first five minutes of the start of the class period. Other assignments will need to be submitted digitally to Moodle. In case Moodle does not work, assignments need to be submitted by email to your instructor AND teaching assistant by the deadline date and time. In general, there is no late policy; if you submit an assignment after it is due, you get zero points for your assignment. In cases of illness or family emergency, you must inform your instructor immediately if you believe you will not be able to submit your assignment on time. In such cases, an exception may be made at the discretion of your instructor.

Classroom Behavior

You are expected to act respectfully towards your fellow classmates, TAs, and instructors inside and outside of the classroom. We have a limited amount of time to cover a lot of material this semester, so you need to pay attention in class, and do your in-class work when it is assigned. Talking on your phone, texting, chatting online, browsing Facebook or other social media sites, and talking excessively with your neighbors about non-class related stuff are just a few examples of behavior that is not acceptable, and will negatively impact your grade.

Grade Policy

The passing grade for the course is D and above. As per CS department policy, students with failing marks within 5% of the passing threshold do not have the option of receiving an Additional Assessment Task at the end of the semester to increase their grade.

Supporting Documents

Course Approval Date

School Learning and Teaching Committee Approval:

Course Origination Date: 2026-04-20

Effective from: Spring 2018